

Diverse Evidence, Better Forecasts: Multi-Agent Deliberation Under Information Asymmetry

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Abstract

Multi-agent systems are increasingly used for forecasting future events, on the assumption that deliberation among multiple LLMs improves reasoning and calibration. Existing approaches, however, overlook a basic design choice: *what information each agent receives*. When all agents observe identical evidence, deliberation collapses into herding rather than genuine belief revision, and the ensemble offers little over a single agent. We propose *designed information asymmetry*: partitioning the evidence into a shared public pool and disjoint private subsets, so that each agent holds exclusive knowledge that can reach the others only through deliberation. We show theoretically that this partition reduces inter-agent error correlation, and instantiate it in **InfoDelphi**, a framework combining relevance-aware evidence routing, rationale-based iterative deliberation, and a calibrated aggregation layer that counteracts systematic LLM overconfidence via asymmetric shrinkage. On POLYGYM, a controlled benchmark of 250 resolved Polymarket questions with fixed pre-retrieved evidence, and on 231 questions from FutureX, InfoDelphi attains Brier scores of 0.158 and 0.201, outperforming nine single-agent and multi-agent baselines under a paired bootstrap test on both datasets, with a 13–18% relative Brier improvement over the strongest baseline. Analysis shows that deliberation under homogeneous input collapses forecast diversity while partitioned evidence preserves it, that the affirmative-side overconfidence shared by every baseline is corrected only by our aggregation layer, and that the full framework keeps its advantage over all baselines when ported to a different backbone model without re-tuning.

1 Introduction

Prediction markets such as Polymarket¹ aggregate the beliefs of thousands of participants into

¹<https://polymarket.com/>

real-time probability estimates for future events, achieving forecasting accuracy that often rivals or exceeds expert panels (Tetlock, 2005). With every question resolved against a definitive ground truth, these markets provide a natural benchmark for machine intelligence: *can an automated system reason over evidence and match human-level forecasting?* Recent work has shown that large language models (LLMs) equipped with retrieved evidence can approach human-level performance on binary forecasting tasks (Halawi et al., 2024; Turtel et al., 2025), but these advances rely primarily on single-agent, single-pass pipelines that treat the entire evidence corpus as a monolithic context window. A natural extension is the *multi-agent* setting, where a panel of LLMs deliberates over the evidence before producing a collective forecast. Such frameworks draw inspiration from well-studied human mechanisms for aggregating distributed knowledge—expert committees, structured interviews, and the Delphi method (Dalkey, 1969)—and have shown promise in general reasoning tasks (Chen et al., 2024; Wang et al., 2024).

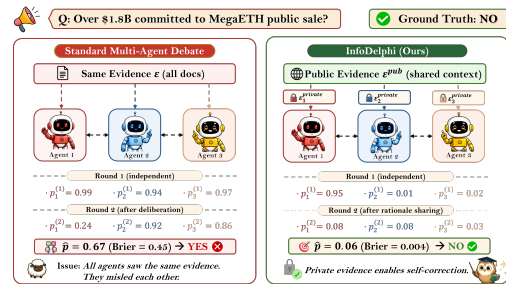


Figure 1: Effect of information asymmetry on a real prediction. **Left:** agents with identical evidence converge to the same incorrect forecast. **Right:** agents with complementary private evidence self-correct through rationale sharing.

Effective collective reasoning, however, requires that participants hold *informationally diverse* views. As formalized in the Condorcet jury tradition and established empirically for hu-

man forecasters (Lorenz et al., 2011), when agents share identical information, deliberation fails to improve collective accuracy: they reach the same conclusions independently, and iterative exchange merely reinforces the shared prior. In LLM systems this problem runs deeper. Recent work shows that models from different providers make correlated errors at rates above 60% (Kim et al., 2025), and that standard multi-agent debate under homogeneous input behaves as a *martingale* whose expected accuracy does not improve over rounds (Choi et al., 2026). Figure 1 illustrates this failure mode on a real prediction market question: when all agents receive identical evidence, they converge to the same incorrect forecast (left), whereas agents equipped with complementary private evidence can self-correct through deliberation (right).

To address this, we propose *designed information asymmetry*, treating information asymmetry not as a limitation to overcome but as a principle to exploit. We introduce **InfoDelphi**, a multi-agent forecasting framework that partitions evidence into a shared public pool and disjoint private subsets, giving each agent distinct expertise while preserving common ground for communication. This mirrors effective human expert panels, where participants contribute complementary knowledge rather than reading the same briefing materials. InfoDelphi constructs informative evidence partitions using BM25 relevance ranking; during deliberation, agents exchange rationale excerpts rather than raw evidence, enabling efficient propagation of private signals. Beyond information structure, we find that LLM forecasters are systematically overconfident on the affirmative side of low-base-rate questions, and that this bias is best corrected *after* deliberation: the final aggregation layer applies an asymmetric shrinkage toward a NO-leaning prior, with parameters validated across datasets.

We evaluate on POLYGYM, a controlled benchmark of 250 resolved binary Polymarket questions with fixed, pre-retrieved evidence shared across all methods, and on 231 resolved questions from the FutureX benchmark (Zeng et al., 2025). Against nine single-agent and multi-agent baselines run with the same backbone model and the same evidence, InfoDelphi achieves the best Brier score on both datasets (0.158 and 0.201), and a paired bootstrap over questions confirms that every pairwise gap is significant at the 95% level. Our analysis

further traces *why* the baselines fall short: deliberation under homogeneous input collapses forecast diversity (herding), and the affirmative-side overconfidence that afflicts every baseline is corrected by our aggregation layer with parameters that transfer across datasets — an advantage that persists even when the same correction is granted to all baselines, and when the framework is ported to a different backbone model.

Our main contributions are as follows:

- We propose *designed information asymmetry* for multi-agent forecasting, and theoretically show that public/private evidence partitioning reduces inter-agent error correlation.
- We introduce **InfoDelphi**, which instantiates this principle through relevance-aware evidence routing, rationale-based deliberation, and a calibrated shrinkage aggregation layer whose parameters transfer across datasets.
- On POLYGYM-250 and FutureX-231, InfoDelphi outperforms nine baselines with paired-bootstrap significance; ablations attribute the gains to information asymmetry and the aggregation layer, and deliberation-dynamics analysis provides direct empirical evidence of herding under homogeneous input.

2 Related Work

LLM-based Forecasting. Early work established a substantial gap between LLM and human forecasting accuracy. Zou et al. (2022) find that language models achieve only 65% accuracy on Autocast compared to 92% for human aggregates, while Schoenegger and Park (2023) show that GPT-4 without augmentation does not significantly outperform a random baseline on Metaculus. Retrieval-augmented generation (Lewis et al., 2020) substantially closes this gap: Yan et al. (2023) improve Autocast accuracy through zero-shot re-ranking and summarization of retrieved news, and Halawi et al. (2024) present a full pipeline achieving near-human performance on Polymarket and Metaculus. More recent work focuses on training: Turtel et al. (2025) fine-tune on 12,100 resolved Polymarket questions, and Chandak et al. (2025) train an 8B model via reinforcement learning that matches proprietary LLMs on prediction benchmarks. On the aggregation side, Schoenegger et al. (2024) show that ensembling diverse LLMs produces forecasts statistically in-

distinguishable from human forecasters, and Tian et al. (2023) document that verbalized LLM probabilities are systematically miscalibrated, motivating post-hoc correction. Benchmark design has matured in parallel: ForecastBench (Karger et al., 2024) and FutureX (Zeng et al., 2025) provide continuously updated evaluations with strict temporal controls, and Paleka et al. (2025) identify systematic pitfalls including temporal leakage and piggybacking on existing forecasts. Despite this progress, existing pipelines treat forecasting as a single-agent process and do not study evidence distribution across a deliberating panel.

Multi-Agent Reasoning and Deliberation.

Multi-agent debate has emerged as a general strategy for improving LLM reasoning (Du et al., 2024). Wang et al. (2022) show that sampling diverse chain-of-thought paths and selecting by majority vote improves accuracy, while Wang et al. (2024) propose Mixture-of-Agents, a layered architecture where later agents condition on all prior outputs. Chen et al. (2024) demonstrate that a round-table conference of diverse LLMs with confidence-weighted voting outperforms single-agent baselines across reasoning benchmarks. However, recent theory suggests multi-agent debate is not uniformly beneficial. Choi et al. (2026) prove that under a Dirichlet-Categorical belief model, standard multi-agent debate forms a *martingale*: expected correctness does not improve over rounds when agents receive identical inputs. Shin (2026) formalize this via the Data Processing Inequality: closed-system deliberation forms a Markov chain, so mutual information with the ground truth can only decrease. Kim et al. (2025) document that LLM errors are 60%+ correlated, implying that naive ensembling hits a non-zero error floor. Our work addresses these limitations by varying the *information structure* of debate rather than its protocol: designed information asymmetry breaks error correlation and converts the martingale into productive belief revision.

Information Aggregation and Collective Intelligence. The conditions under which groups outperform individuals are well studied. Surowiecki (2004) identifies independence, diversity, and decentralized knowledge as key conditions for collective intelligence. The DeGroot model (DeGroot, 1974) formalizes iterative belief aggregation, and the Delphi method (Dalkey, 1969) ap-

plies structured multi-round elicitation to reduce expert disagreement. For probabilistic forecasting, the linear opinion pool (Stone, 1961) and logit-space extremization (Satopää et al., 2014) provide principled aggregation; shrinkage toward a prior is the classical remedy when individual forecasts are overconfident. Meanwhile, social influence can suppress diversity without improving accuracy (Lorenz et al., 2011), and informational cascades may cause agents to ignore private signals once consensus emerges (Bikhchandani et al., 1992). Our framework operationalizes the positive conditions for collective intelligence while explicitly mitigating herding and information loss through evidence partitioning, rationale-based communication, and calibrated aggregation.

3 Methodology

3.1 Problem Formulation

Let q denote a binary forecasting question with resolution date τ and ground truth $y \in \{0, 1\}$. Each question is accompanied by a retrieved evidence corpus $\mathcal{E} = \{e_1, \dots, e_n\}$, where e_i consists of a title, URL, and textual snippet retrieved prior to τ . Given the question q and evidence corpus $\mathcal{E}$, the goal is to estimate $p(y = 1 \mid q, \mathcal{E})$, where $p \in [0, 1]$ denotes the predicted probability that the event resolves positively.

3.2 Theoretical Motivation

We begin by formalizing why information asymmetry improves collective forecasting, drawing on the bias-variance-covariance decomposition of the ensemble mean squared error.

Homogeneous input induces correlated errors.

Consider J agents each receiving the same evidence $\mathcal{E}$. Let p_j denote agent j 's forecast and y the ground truth. The ensemble forecast $\bar{p} = \frac{1}{J} \sum_j p_j$ has mean squared error:

$$\text{MSE}(\bar{p}) = \text{Bias}^2 + \bar{\sigma}^2 - \text{Div}, \quad (1)$$

where $\bar{\sigma}^2 = \frac{1}{J} \sum_j \text{Var}(p_j)$ is the average agent variance and $\text{Div} = \frac{1}{J} \sum_j \text{Var}(p_j - \bar{p})$ is the diversity term (Wood et al., 2023). When all agents observe identical input, their prediction errors $\epsilon_j = p_j - y$ are driven by the same evidence and thus highly correlated: $\text{Cov}(\epsilon_i, \epsilon_j) \approx \text{Var}(\epsilon)$ for $i \neq j$. In this regime the diversity term collapses toward zero, and the ensemble provides no benefit over

a single agent. Recent empirical work confirms this: LLM errors are correlated across providers, and optimal aggregation under full correlation has a non-zero error floor (Kim et al., 2025; Turkmen et al., 2026).

Information asymmetry reduces error correlation. We now show that a public/private evidence partition directly reduces this correlation.

Proposition 3.1 (Diversity-Induced Decorrelation). *Let each agent’s prediction error decompose as $\epsilon_j = \rho \cdot \epsilon^{\text{pub}} + (1 - \rho) \cdot \epsilon_j^{\text{priv}}$, where ϵ^{pub} is the error component driven by shared public evidence, ϵ_j^{priv} is the component driven by agent j ’s private evidence, and $\rho \in [0, 1]$ is the public ratio. Let $\sigma_{\text{pub}}^2 = \text{Var}(\epsilon^{\text{pub}})$ and assume:*

- (i) $\text{Cov}(\epsilon_i^{\text{priv}}, \epsilon_j^{\text{priv}}) = 0$ for $i \neq j$ (private-private independence);
- (ii) $\text{Cov}(\epsilon^{\text{pub}}, \epsilon_k^{\text{priv}}) = 0$ for all k (public-private independence);
- (iii) $\text{Var}(\epsilon_j^{\text{priv}}) = \sigma_{\text{priv}}^2$ for all j (symmetric private noise).

Then:

$$\text{Corr}(\epsilon_i, \epsilon_j) = \frac{\rho^2 \sigma_{\text{pub}}^2}{\rho^2 \sigma_{\text{pub}}^2 + (1-\rho)^2 \sigma_{\text{priv}}^2}. \quad (2)$$

Proof. By bilinearity of covariance and assumptions (i)–(ii), all cross-terms vanish, giving $\text{Cov}(\epsilon_i, \epsilon_j) = \rho^2 \sigma_{\text{pub}}^2$. By assumptions (ii)–(iii), the cross-term in the variance expansion also vanishes, so $\text{Var}(\epsilon_j) = \rho^2 \sigma_{\text{pub}}^2 + (1-\rho)^2 \sigma_{\text{priv}}^2$, which equals $\text{Var}(\epsilon_i)$ by symmetry. Dividing covariance by $\sqrt{\text{Var}(\epsilon_i)\text{Var}(\epsilon_j)} = \text{Var}(\epsilon_j)$ gives (2). $\square$

Proposition 3.1 shows that inter-agent error correlation decreases monotonically as ρ decreases from 1 to 0. As a concrete instance, at the default $\rho = 0.5$ with equal-variance components ($\sigma_{\text{pub}} = \sigma_{\text{priv}}$), correlation equals $0.25/(0.25 + 0.25) = 0.5$, half the all-public baseline ($\rho=1$, correlation = 1). At $\rho = 0$ (all-private), agents are fully decorrelated; at $\rho = 1$ (all-public, standard debate), correlation reaches its maximum.

Corollary 3.1.1 (Optimal Public Ratio Trade-off). *Decreasing ρ reduces error correlation (improving ensemble gain) but also reduces the shared context necessary for agents to interpret each other’s rationales. An optimal $\rho^* \in (0, 1)$ therefore exists that balances diversity against communicability.*

Rationale sharing enables cross-agent information transfer. Even with disjoint evidence, agents cannot benefit from each other’s private signals unless they communicate their *reasoning*, not just their conclusions. By the Data Processing Inequality (Shin, 2026), a deliberation system where agents only exchange numeric estimates forms a Markov chain $\mathcal{E}_j \rightarrow p_j^{(1)} \rightarrow p_j^{(2)} \rightarrow \dots$, so $I(y; p_j^{(r+1)}) \leq I(y; p_j^{(r)})$: information about the ground truth can only decrease. Sharing rationales breaks this Markov structure: agent i now observes a summary of agent j ’s reasoning over $\mathcal{E}_j^{\text{priv}}$, effectively receiving a lossy projection of j ’s private evidence. This converts the system from closed to open, allowing $I(y; p_j^{(r+1)}) > I(y; p_j^{(r)})$, which is a necessary condition for iterative deliberation to improve over a single round.

3.3 The InfoDelphi Framework

Overview. Figure 2 illustrates the end-to-end InfoDelphi pipeline, which instantiates the three conditions identified above — diverse signals, common ground, and exchanged reasoning — through three corresponding stages: (1) *evidence routing* partitions $\mathcal{E}$ into a shared public pool and disjoint private subsets via BM25 relevance ranking; (2) *iterative deliberation* enables agents to exchange rationale excerpts across rounds, converting private signals into shared knowledge; (3) *calibrated aggregation* averages final-round forecasts and applies an asymmetric shrinkage that corrects systematic overconfidence.

Evidence Routing. We consider a panel of J agents $\{A_1, \dots, A_J\}$ and a public ratio $\rho \in [0, 1]$. The corpus is partitioned into a shared public subset $\mathcal{E}^{\text{pub}}$ and J disjoint private subsets:

$$\mathcal{E} = \mathcal{E}^{\text{pub}} \cup \bigsqcup_{j=1}^J \mathcal{E}_j^{\text{priv}}, \quad |\mathcal{E}^{\text{pub}}| = \lfloor \rho \cdot |\mathcal{E}| \rfloor. \quad (3)$$

Agent A_j observes the private view $\mathcal{E}_j = \mathcal{E}^{\text{pub}} \cup \mathcal{E}_j^{\text{priv}}$. To maximize information diversity across private subsets, we allocate documents via *BM25 relevance routing*: each document e_i receives a relevance score $s_i = \text{BM25}(e_i, q)$. The top- $|\mathcal{E}^{\text{pub}}|$ scoring documents form $\mathcal{E}^{\text{pub}}$: since these are the most directly relevant to the question, they serve as the shared “language” through which agents can interpret each other’s reasoning during deliberation. The remaining documents are sorted by

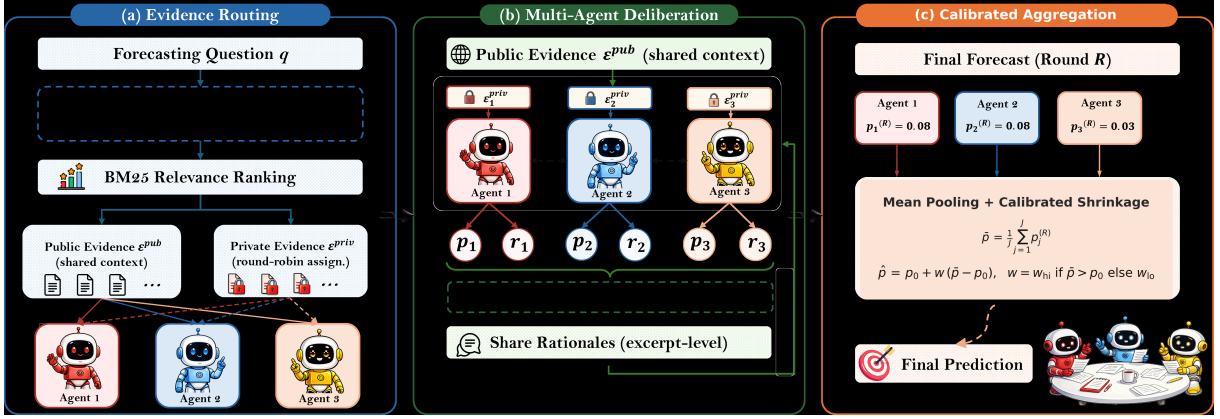


Figure 2: Overview of the **InfoDelphi** framework. (a) **Evidence Routing**: documents are ranked by BM25 relevance and distributed round-robin, constructing a shared public pool and disjoint private subsets for each agent. (b) **Multi-Agent Deliberation**: agents exchange structured forecasts and rationales over two rounds, enabling private information to propagate. (c) **Calibrated Aggregation**: final-round forecasts are averaged and passed through an asymmetric shrinkage transform toward a NO-leaning prior p_0 , correcting systematic overconfidence.

s_i and assigned to agents in round-robin order, so that adjacent relevance levels are spread across distinct private views.

Iterative Deliberation with Rationale Sharing.

At each round $r \in \{1, \dots, R\}$, agent A_j produces a structured forecast

$$f_j^{(r)} = (p_j^{(r)}, \ell_j^{(r)}, r_j^{(r)}), \quad (4)$$

where $p_j^{(r)} \in [0, 1]$ is a probability estimate, $\ell_j^{(r)} \in \{0, 1\}$ is a categorical prediction, and $r_j^{(r)}$ is a free-text rationale. The agent conditions on its private view and a history summary of prior-round forecasts:

$$f_j^{(r)} \sim \text{LLM}(q, \mathcal{E}_j, h^{(r-1)}), \quad (5)$$

where $h^{(0)} = \emptyset$ and, for $r \geq 2$, $h^{(r-1)} = \{(p_k^{(r-1)}, \ell_k^{(r-1)}, r_k^{(r-1)})\}_{k \neq j}$. Crucially, $h^{(r-1)}$ includes rationale excerpts from other agents, not only their numeric estimates; agents may also quote the specific evidence supporting an argument (evidence pooling), so that a private document cited by one agent becomes visible context for the others. The forecasting prompt instructs agents to reason about base rates and resolution criteria before committing to a probability (the *calibrated prompt*; Appendix J). We study two deliberation protocols: *independent forecasting*, where agents reason solely from their private view (one round), and *iterative deliberation* (the full InfoDelphi protocol), where agents additionally observe all other agents' Round 1 forecasts and rationales before producing their final prediction.

Calibrated Aggregation. Given the final-round forecasts $\{p_j^{(R)}\}_{j=1}^J$, InfoDelphi first pools them by the mean, $\bar{p} = \frac{1}{J} \sum_j p_j^{(R)}$, and then applies an *asymmetric shrinkage* toward a prior p_0 :

$$\hat{p} = p_0 + w(\bar{p} - p_0), \quad w = \begin{cases} w_{\text{hi}} & \bar{p} > p_0, \\ w_{\text{lo}} & \bar{p} \leq p_0, \end{cases} \quad (6)$$

with $p_0 = 0.30$, $w_{\text{lo}} = 0.8$, and $w_{\text{hi}} = 0.5$. The design responds to a systematic bias we observe across prompts and datasets: LLM forecasters are overconfident specifically on the *affirmative* side of low-base-rate questions, so forecasts above the prior are shrunk harder ($w_{\text{hi}} < w_{\text{lo}}$) than forecasts below it. The three parameters are fit on one dataset and validated on the other (Section 6.2); the same setting transfers between Polymarket and FutureX without re-tuning. Section 6.2 contrasts this aggregation-layer correction with prompt-level attempts at instilling skepticism, which depress the entire forecast distribution rather than the overconfident region.

4 Experimental Setup

Datasets. Current LLM forecasting systems (Halawi et al., 2024; Turtel et al., 2025) primarily emphasize *retrieval quality*, while an equally important capability is *information utilization*: how agents interpret, distribute, and collaboratively reason over retrieved evidence. To study this dimension in isolation, we construct **POLYGYM**, a controlled forecasting benchmark of 250 resolved binary Polymarket questions (resolution dates 2025-09 to 2025-12; empirical

YES rate 0.232) spanning Sports, Business, Politics, and other domains. Each question is paired with up to 20 pre-retrieved documents collected before market resolution under a strict temporal cutoff, and all methods operate over the same fixed evidence pool, eliminating variability from changing web environments and unstable search results (details in Appendix B). To test generalization beyond Polymarket, we additionally evaluate on 231 resolved questions from the FutureX benchmark (Zeng et al., 2025) (YES rate 0.355), constructed with the same fixed-evidence protocol.

Baselines. We compare against two categories of baselines, all run with the same backbone model and the same evidence pools (details in Appendix E). *Single-agent methods:* Chain-of-Thought (Wei et al., 2022), Self-Consistency (Wang et al., 2022), Sequential Bayesian (Shi et al., 2023), Superforecaster (Karger et al., 2024), and the structured scratchpad pipeline of Halawi et al. (Halawi et al., 2024). *Multi-agent methods with homogeneous input:* Standard Debate (Du et al., 2024), MoA (Wang et al., 2024), Crowd Ensemble (Schoenegger et al., 2024), and AIA Forecaster (Alur et al., 2025). Notably, our Standard Debate implementation shares InfoDelphi’s prompt, deliberation protocol, and mean aggregation, differing only in information structure ($\rho=1$ with full evidence sharing); it therefore doubles as a controlled ablation of information asymmetry.

Metrics and significance. We report the *Brier score* $BS = \frac{1}{N} \sum_{k=1}^N (\hat{p}_k - y_k)^2$ (lower is better), *accuracy*, the fraction of questions for which $1[\hat{p} \geq 0.5]$ matches y , and *AUC*, the probability that a randomly chosen YES question receives a higher forecast than a randomly chosen NO question. AUC is invariant to any strictly monotone recalibration — including the shrinkage of Eq. (6) — and therefore measures ranking skill independently of the calibration layer. For every pairwise comparison we run a paired bootstrap over questions (5,000 resamples, fixed seed) and report the 95% confidence interval of the mean Brier difference; a gap is called significant when the interval excludes zero.

Implementation. All experiments use gpt-5.4-mini via the OpenRouter API² as the

primary model; Table 2 additionally evaluates the full framework on gemini-3.1-flash-lite and gpt-4o-mini. We use $J = 3$ agents, $R = 2$ deliberation rounds, $\rho = 0.5$, temperature 0.7, and the shrinkage parameters of Eq. (6) for all main results.

5 Main Results

Table 1 compares InfoDelphi against all nine baselines on both datasets.

Overall performance. InfoDelphi attains the best Brier score on both benchmarks (0.1575 on POLYGYM-250, 0.2006 on FutureX-231), and the improvement over *every* baseline is significant under the paired bootstrap on both datasets (all 95% CIs exclude zero; Appendix Table 8). Relative to the strongest baseline (Standard Debate), Brier falls by 13.2% and 17.6%, respectively. The gains come from two designed mechanisms working in sequence: *evidence routing* gives each agent a partially private view of the corpus, creating the decorrelated signals that Proposition 3.1 requires, and *rationale-based deliberation* lets agents propagate those private signals to the panel. Importantly, the raw ensemble without the shrinkage layer (0.1647 / 0.2212) already outperforms every baseline on both datasets, so the ranking is established by the information structure alone; calibrated aggregation then widens the margin. AUC makes the same point in calibration-free terms: because AUC is unaffected by the shrinkage transform, InfoDelphi’s 0.757 / 0.711 — highest in every column against 0.729 / 0.694 for the best baselines — reflects pure ranking skill that no recalibration of a baseline could recover.

Information diversity, not agent count, drives multi-agent gains. The homogeneous-input baselines deploy the same or greater inference budget than InfoDelphi — multiple agents, multiple rounds, and varied aggregation schemes — yet on FutureX every one of them except Standard Debate is *worse* than the best single-agent method (MoA 0.2944 and Crowd Ensemble 0.2888 vs. 0.2596 for the single Superforecaster prompt), and the same pattern appears in weaker form on POLYGYM. When agents read identical evidence, additional agents and rounds mostly redistribute the same information, consistent with the martingale view of homogeneous debate (Choi et al., 2026) and with Proposition 3.1: without

²openrouter.ai

Table 1: Main results on POLYGYM-250 and FutureX-231. All methods use gpt-5.4-mini over the same fixed pre-resolution evidence pools. AUC is invariant to monotone recalibration, so it isolates ranking skill from the calibration layer (identical for our raw and full rows). †: InfoDelphi (full) is significantly better under a paired bootstrap over questions (5,000 resamples; 95% CI of the Brier difference excludes 0; Appendix Table 8). Best in **bold**.

Method	POLYGYM-250			FutureX-231		
	Brier ↓	Acc ↑	AUC ↑	Brier ↓	Acc ↑	AUC ↑
<i>Single-agent</i>						
Chain-of-Thought (Wei et al., 2022)†	0.2063	0.696	0.678	0.2611	0.606	0.663
Self-Consistency (Wang et al., 2022)†	0.1965	0.712	0.699	0.2713	0.550	0.630
Sequential Bayesian (Shi et al., 2023)†	0.2523	0.540	0.591	0.3043	0.485	0.534
Superforecaster (Karger et al., 2024)†	0.1937	0.700	0.722	0.2596	0.636	0.672
Halawi et al. (Halawi et al., 2024)†	0.1944	0.716	0.711	0.2654	0.567	0.629
<i>Multi-agent, homogeneous input</i>						
Standard Debate (Du et al., 2024)†	0.1815	0.720	0.711	0.2433	0.602	0.694
MoA (Wang et al., 2024)†	0.1905	0.720	0.716	0.2944	0.545	0.599
Crowd Ensemble (Schoenegger et al., 2024)†	0.1992	0.700	0.700	0.2888	0.524	0.599
AIA Forecaster (Alur et al., 2025)†	0.1897	0.704	0.729	0.2764	0.550	0.635
<i>Ours</i>						
InfoDelphi w/o calibrated shrink	0.1647	0.748	0.757	0.2212	0.688	0.711
InfoDelphi (full)	0.1575	0.744	0.757	0.2006	0.701	0.711

asymmetry, inter-agent error correlation stays high and aggregation yields diminishing returns. Standard Debate is the partial exception — iterative belief revision does help — but agents sharing one evidence view can only reinforce existing interpretations, capping it well below InfoDelphi.

Affirmative overconfidence is universal among baselines; InfoDelphi attacks it twice. Across all nine baselines, forecasts of $p \geq 0.6$ resolve YES far less often than stated: stated confidence averages 0.71–0.75 against hit rates of 0.28–0.38 on POLYGYM, and 0.76–0.80 against 0.32–0.51 on FutureX. Sequential Bayesian, whose step-wise updates compound this drift, is the weakest method on both datasets (0.2523 / 0.3043). InfoDelphi counters this shared failure mode at two designed levels: deliberation under partitioned evidence sharply reduces *exposure* to the overconfident region (14 vs. 28–55 high-confidence forecasts on POLYGYM; 69 vs. 84–107 on FutureX), and the asymmetric shrinkage of Eq. (6) corrects what remains (0.1647 $\rightarrow$ 0.1575 and 0.2212 $\rightarrow$ 0.2006). No baseline has a mechanism for either effect; Section 6.2 analyzes both, and Appendix F shows that even granting every baseline the same shrinkage post hoc leaves InfoDelphi best on both datasets.

Gains transfer across backbones. To verify that the advantage stems from the infor-

Table 2: Cross-model generalization: the full InfoDelphi configuration (same routing, deliberation, and shrinkage parameters, no re-tuning) on a different vendor’s backbone, vs. the strongest baseline. The Gemini Polymarket run covers 248/250 questions.

	POLYGYM		FutureX	
	Brier ↓	Acc ↑	Brier ↓	Acc ↑
<i>Strongest baseline (Std. Debate)</i>	0.1815	0.720	0.2433	0.602
InfoDelphi, gpt-5.4-mini	0.1575	0.744	0.2006	0.701
InfoDelphi, gemini-3.1-flash-lite	0.1624	0.758	0.2040	0.635

mation structure rather than one model’s idiosyncrasies, we run the full InfoDelphi configuration — identical routing, deliberation protocol, and shrinkage parameters, with no re-tuning — on gemini-3.1-flash-lite (Table 2). The Gemini instantiation (0.1624 / 0.2040) still beats all nine baselines on both datasets, even though every baseline runs on the stronger gpt-5.4-mini; its independent-forecast round alone (0.157 / 0.203) already clears the strongest baseline, confirming that evidence partitioning provides value before any deliberation occurs. A gpt-4o-mini run on FutureX shows the same ordering (0.2370 vs. 0.2212 for gpt-5.4-mini, both without shrinkage).

Table 3: Component ablation (Brier $\downarrow$). Each row removes one component from the full system. “— asymmetry” gives every agent the full evidence pool ($\rho=1$, full sharing) with the same calibrated prompt, i.e., the Standard Debate configuration of our pipeline. FutureX round-1 forecasts are recovered from the winner run’s recorded round-0 agent outputs. Paired 95% CIs for each removal are given in the text.

Configuration	POLYGYM	FutureX
InfoDelphi (full)	0.1575	0.2006
— calibrated shrink (mean agg.)	0.1647	0.2212
— deliberation (round 1 only)	0.1698	0.2352
— information asymmetry	0.1815	0.2433

6 Analysis

6.1 Why Information Asymmetry Beats Homogeneous Debate

Creating asymmetry is the largest single factor. Table 3 removes one designed component at a time. The biggest degradation comes from removing information asymmetry — giving every agent the full pool with full sharing, i.e., exactly the Standard Debate configuration with our prompt and aggregation: Brier worsens from 0.1647 to 0.1815 on POLYGYM (paired $\Delta = -0.0169$, CI $[-0.0314, -0.0029]$) and from 0.2212 to 0.2433 on FutureX ($\Delta = -0.0221$, CI $[-0.0515, +0.0081]$), significant on POLYGYM and directionally consistent on FutureX. Because this comparison holds the prompt, protocol, and aggregation fixed, the gap over the strongest baseline is attributable to information structure alone. Removing deliberation (round-1 forecasts only) costs 0.005 on POLYGYM and 0.014 on FutureX ($\Delta = -0.0141$, CI $[-0.0325, +0.0027]$), and removing the shrinkage layer costs 0.007 / 0.021: partitioning creates the *potential* for improvement, deliberation and calibrated aggregation *realize* it.

The advantage rests on the principle, not on a tuned hyperparameter. Table 4 varies every remaining design dimension one at a time. The default configuration is highly stable (Brier std 0.0002 after shrinkage across 3 seeds), and against this yardstick every variation — $\rho \in \{0.3, 0.7\}$, a third deliberation round, numeric-only sharing, random routing — shifts Brier by at most 0.004, within seed noise (all paired CIs vs. the default straddle zero). The single catastrophic choice is $\rho=1$: full homogenization degrades Brier by 0.017 and AUC from 0.757 to 0.711, the only signifi-

Table 4: Design-space robustness on POLYGYM-250. Each row changes one dimension of the default configuration ($R=2$, $\rho=0.5$, rationale sharing, BM25 routing). The default is stable across 3 seeds (Brier 0.1643 ± 0.0011 raw, 0.1577 ± 0.0002 +shrink); every variation except full homogenization ($\rho=1$) lies within seed noise ($\dagger$: paired 95% CI vs. default excludes 0, raw comparison).

Setting	Brier (raw)	+shrink	AUC
Default	0.1647	0.1575	0.757
<i>Deliberation rounds</i>			
$R=1$ (independent)	0.1698	0.1596	0.744
$R=3$	0.1593	0.1544	0.768
<i>Public ratio</i>			
$\rho=0.3$	0.1594	0.1553	0.763
$\rho=0.7$	0.1561	0.1536	0.770
$\rho=1.0$ (homogeneous) †	0.1815	0.1650	0.711
<i>Rationale sharing</i>			
numeric estimates only	0.1687	0.1585	0.751
<i>Evidence routing</i>			
random split	0.1665	0.1586	0.746

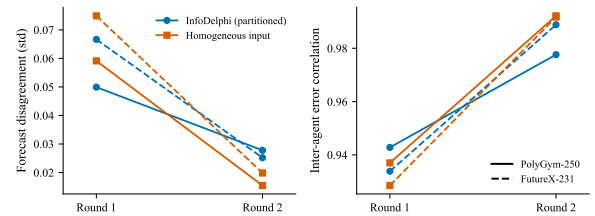


Figure 3: Deliberation dynamics with and without information asymmetry. Left: mean per-question standard deviation of agent forecasts; right: mean pairwise inter-agent error correlation. Deliberation under homogeneous input collapses forecast diversity (herding); partitioned evidence preserves independent signal after deliberation on both datasets.

cant change in the grid. The component directions still agree with the theory — a third round adds nothing once private signals have propagated ($R=3 \approx R=2$), and replacing rationale sharing or relevance routing lowers AUC (0.751 / 0.746 vs. 0.757) — but their magnitudes are secondary. What separates InfoDelphi from the baselines is therefore not a fragile hyperparameter setting but the presence of private evidence at all: any evidence partition beats no partition, and no setting of the other knobs rescues the homogeneous configuration.

Homogeneous deliberation herds; partitioned deliberation preserves diversity. Figure 3 traces *why* the strongest multi-agent baseline loses to InfoDelphi despite an identical protocol. In round 1 the two information structures

show comparable per-question disagreement. After one round of deliberation, however, disagreement under homogeneous input collapses by 74–79% (e.g., $0.059 \rightarrow 0.015$ on POLYGYM) and mean pairwise error correlation rises to 0.99: the panel has effectively merged into a single agent, and averaging its members can no longer cancel errors. With partitioned evidence, agents converge far less ($0.050 \rightarrow 0.028$) and retain lower error correlation (0.978), preserving the diversity term of Eq. (1). This is direct empirical evidence for the mechanism of Proposition 3.1: private evidence gives each agent an independent basis for conviction that social pressure alone cannot override.

Asymmetry makes agents confident about different things. A natural worry is that withholding evidence merely makes agents timid. The opposite holds: after the final round, InfoDelphi’s agents are as extreme as Standard Debate’s (mean $|p - 0.5|$ of 0.277 vs. 0.271 on POLYGYM), but their inter-agent variance is $2.7\times$ higher ($1.3\times$ on FutureX). Information asymmetry does not reduce individual confidence — it decorrelates its *direction*, so the mean-pooling step of Section 3.3 averages genuinely complementary judgments rather than three copies of one belief. This validates the evidence-routing design directly: the value of the partition lies in the diversity it protects.

6.2 Calibration: A Shared Failure Mode, Corrected in Aggregation

The miscalibration is one-sided, motivating an asymmetric correction. Section 5 showed every baseline overshooting on high-confidence YES forecasts. Our raw ensemble inherits the same model-level bias (Tian et al., 2023), but only on one side: forecasts below the prior ($p < 0.3$) and in the mid-band are close to calibrated (e.g., stated 0.42 vs. empirical 0.45 on POLYGYM), while the affirmative band overshoots badly (stated 0.79 vs. hit rate 0.29 on POLYGYM; 0.81 vs. 0.59 on FutureX). This one-sidedness is precisely why Eq. (6) shrinks asymmetrically ($w_{hi} < w_{lo}$): the correction acts strongly where the errors live and leaves the calibrated region nearly untouched. Symmetric alternatives — shrinking everything toward 0.5, or prompt-level skepticism instructions — depress the whole forecast distribution and can underperform a constant base-rate predictor (Appendix G).

The correction transfers because the bias is model-level, not dataset-level. The three shrinkage parameters are fit on Polymarket and applied to FutureX unchanged, improving Brier from 0.2212 to 0.2006 and accuracy from 0.688 to 0.701; re-fitting on FutureX recovers the same setting. A correction that transfers across question distributions with different base rates (0.232 vs. 0.355) is capturing a property of the forecaster, not an artifact of one dataset. We also verify the comparison is not an artifact of the calibration layer itself: applying the identical transform to every baseline post hoc (Appendix F) helps them too, but InfoDelphi remains the best system on both datasets — the residual advantage is exactly the part that calibration cannot supply, namely the decorrelated information structure of Section 6.1. A companion study of the Sports domain (Appendix H) probes the boundary of this advantage: information asymmetry improves the *utilization* of informative evidence, but cannot create information the corpus does not contain.

7 Conclusion

We presented **InfoDelphi**, a multi-agent forecasting framework built on a simple but overlooked insight: collective reasoning only outperforms individual reasoning when participants hold genuinely different information. Treating information asymmetry as a first-class design principle — and correcting overconfidence in the aggregation layer rather than the prompt — yields the best Brier score among ten systems on two fixed-evidence benchmarks, with paired-bootstrap significance against every baseline. Our analysis localizes the mechanism (herding collapse under homogeneous input) and its boundary (evidence-limited domains), suggesting that in multi-agent LLM systems, *what each agent reads* deserves the same design attention as *how agents talk*.

Limitations

Theoretical assumptions. Proposition 3.1 rests on three idealizing assumptions: a linear error decomposition, public–private independence, and symmetric private noise. In practice, documents retrieved for the same question share topical correlations, partially violating independence and making the decorrelation bound approximate; LLMs, as non-linear functions of their inputs, need not satisfy the fixed-weight decomposition. Consis-

tently, the empirical error correlations in Figure 3 remain high (~ 0.94) even with partitioned evidence, because model-level correlation persists regardless of evidence allocation.

Effect sizes and scope. The deliberation-round effect is small relative to question-level noise and not individually significant, and the advantage over the strongest homogeneous-input baseline, while significant on POLYGYM, is directional on FutureX; when the shrinkage transform is granted to all baselines (Appendix F), InfoDelphi remains best on both datasets but several pairwise gaps are no longer individually significant. Our baseline suite runs on a single backbone model (gpt-5.4-mini); the cross-model check of Table 2 covers additional backbones for InfoDelphi only, and our experiments are limited to binary questions and fixed evidence pools — live-retrieval settings, open-ended questions, and heterogeneous model panels remain untested. The shrinkage parameters, while validated across our two datasets, are not guaranteed to transfer to models or question distributions with different bias profiles. API cost scales linearly with agents $\times$ rounds (1,500 forecasting calls for a full POLYGYM-250 run at $J=3$, $R=2$), which limited exploration of larger panels and longer deliberation.

Ethics Considerations

This work presents a forecasting framework and does not directly enable disinformation or harm beyond risks inherent to LLM-based systems generally. However, automated probability estimates generated at scale could in principle be misused to manipulate market sentiment or create false impressions of consensus. We recommend that InfoDelphi be used as a decision-support tool with human oversight in high-stakes deployment contexts.

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A Discussion

Key insight. Our central finding is that *designed information asymmetry* is necessary for effective multi-agent deliberation. When all agents receive identical evidence, iterative discussion provides little measurable benefit, as agents merely reinforce correlated beliefs rather than contribute complementary knowledge (Figure 3). This aligns with the martingale characterization of homogeneous-input debate and suggests that the widely adopted practice of giving all agents the same context fundamentally limits the value of multi-agent interaction.

Future work. Natural extensions include (1) adaptive routing that dynamically adjusts evidence allocation based on emerging disagreement, (2) more expressive routing algorithms beyond BM25, such as dense retrieval or learned semantic partitioning, (3) learned aggregation replacing the fixed shrinkage parameters, (4) evaluation on open-ended and multi-choice forecasting tasks, and (5) heterogeneous model panels that leverage model-level diversity in addition to evidence-level asymmetry.

Potential risks. Designed information asymmetry encourages agents to maintain distinct beliefs, but it may also amplify biased or misleading evidence partitions. If private evidence subsets contain low-quality, adversarial, or outdated documents, agents may become confidently polarized around incorrect conclusions rather than self-correcting through deliberation. In addition, rationale sharing introduces a potential attack surface, where persuasive but flawed rationales propagate errors across agents. Future deployments should incorporate evidence verification, source reliability estimation, and safeguards against adversarial persuasion.

B POLYGYM Benchmark Details

Motivation. Agentic forecasting systems for prediction markets require two distinct capabili-

ties: (1) *retrieval*, identifying relevant information from the web, and (2) *utilization*, reasoning over that information to produce calibrated probability estimates. Existing benchmarks predominantly evaluate these capabilities jointly (Table 5), making it difficult to determine whether performance gains arise from better retrieval or better collective reasoning. Live benchmarks (Karger et al., 2024; Zeng et al., 2025) additionally introduce variability from changing web environments and unstable evidence availability, and retrieval-augmented systems continue to exhibit degradation over time, suggesting that access to information alone does not solve the utilization challenge (Dai et al., 2025). POLYGYM therefore fixes the evidence pool: all methods observe the same pre-retrieved corpus, while different agents may observe different subsets of it. Conceptually, this parallels controlled retrieval settings in QA research (Chen et al., 2021; Yang et al., 2018) that separate retrieval quality from downstream reasoning.

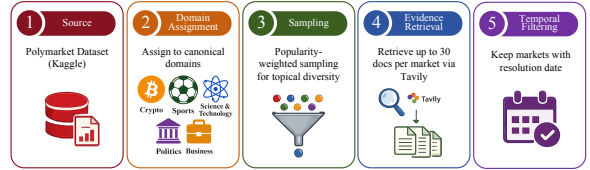


Figure 4: POLYGYM data construction pipeline.

Construction. POLYGYM is derived from a large-scale Polymarket dataset (Semedov, 2025) consisting of approximately 40,000 events and 100,000 prediction markets. Events are assigned to canonical domains using metadata fields, and popularity-weighted sampling ensures topical diversity. After temporal filtering (below), the benchmark contains 250 resolved binary markets: Sports (106), Business (28), Politics (23), Entertainment (6), Geopolitics (5), and other domains (82). The empirical YES rate is 0.232. All questions and evidence documents are in English.

Evidence. For each market, we retrieve up to 20 documents via the Tavily search API³ using queries derived from the question text (mean 19.5 documents per question). A strict temporal cutoff ensures all evidence precedes the market resolution date. Retrieved documents are deduplicated and truncated to 500 characters.

³<https://www.tavily.com/>

Setup	Static	Reproducible	Time Control	Fixed Evidence	Realistic	Agent Study
ForecastBench (Karger et al., 2024)	×	×	✓	×	✓	×
FutureX (Zeng et al., 2025)	×	×	✓	×	✓	×
PolyBench (Cheng et al., 2026)	×	×	✓	×	✓	×
Prediction Arena (Zhang et al., 2026)	×	×	✓	×	✓	✓
Zou et al. (Zou et al., 2022)	✓	×	×	×	×	×
Halawi et al. (Halawi et al., 2024)	✓	×	×	×	×	×
POLYGYM (Ours)	✓	✓	✓	✓	✓	✓

Table 5: Comparison of forecasting evaluation setups. POLYGYM provides fixed evidence and full reproducibility, enabling controlled study of information utilization and multi-agent deliberation.

Temporal filtering. To reduce contamination from LLM prior knowledge, we retain only markets with resolution dates between 2025-09-09 and 2025-12-02, after the primary model’s knowledge cutoff.

Licenses and intended use. The source Poly-market dataset (Semedov, 2025) is licensed under the Open Data Commons Database Contents License (DbCL) v1.0. Evidence documents retrieved via the Tavily Search API are used solely for non-commercial research purposes under its API terms of service and are not independently redistributed. POLYGYM is intended for research use only. The accompanying code is released under the MIT License.

C Proof of Proposition 3.1

We provide the complete derivation. Let $\epsilon_j = \rho \cdot \epsilon^{\text{pub}} + (1 - \rho) \cdot \epsilon_j^{\text{priv}}$ for agent $j \in \{1, \dots, J\}$.

Covariance between agents. For $i \neq j$, expanding by bilinearity of covariance:

$$\begin{aligned} \text{Cov}(\epsilon_i, \epsilon_j) &= \rho^2 \text{Var}(\epsilon^{\text{pub}}) \\ &\quad + \rho(1-\rho) \text{Cov}(\epsilon^{\text{pub}}, \epsilon_j^{\text{priv}}) \\ &\quad + \rho(1-\rho) \text{Cov}(\epsilon_i^{\text{priv}}, \epsilon^{\text{pub}}) \\ &\quad + (1-\rho)^2 \text{Cov}(\epsilon_i^{\text{priv}}, \epsilon_j^{\text{priv}}). \end{aligned} \quad (7)$$

By the independence assumptions, $\text{Cov}(\epsilon_i^{\text{priv}}, \epsilon_j^{\text{priv}}) = 0$ and $\text{Cov}(\epsilon^{\text{pub}}, \epsilon_j^{\text{priv}}) = 0$, so $\text{Cov}(\epsilon_i, \epsilon_j) = \rho^2 \text{Var}(\epsilon^{\text{pub}})$.

Individual variance. Since ϵ^{pub} and ϵ_j^{priv} are independent, the cross term vanishes: $\text{Var}(\epsilon_j) = \rho^2 \text{Var}(\epsilon^{\text{pub}}) + (1-\rho)^2 \text{Var}(\epsilon_j^{\text{priv}})$.

Correlation. Let $\sigma_{\text{pub}}^2 = \text{Var}(\epsilon^{\text{pub}})$ and assume agents are symmetric ($\text{Var}(\epsilon_j^{\text{priv}}) = \sigma_{\text{priv}}^2$ for all

j), which implies $\text{Var}(\epsilon_i) = \text{Var}(\epsilon_j)$ and simplifies the correlation to

$$\text{Corr}(\epsilon_i, \epsilon_j) = \frac{\rho^2 \sigma_{\text{pub}}^2}{\rho^2 \sigma_{\text{pub}}^2 + (1-\rho)^2 \sigma_{\text{priv}}^2}. \quad (8)$$

In the homogeneous baseline where $\rho = 1$: $\text{Corr}(\epsilon_i, \epsilon_j) = 1$. As ρ decreases, correlation decreases monotonically, reaching 0 when $\rho = 0$. At our default $\rho = 0.5$ with equal-variance components, $\text{Corr} = 0.5$, a 50% reduction relative to $\rho = 1$. $\square$

D Proof of Corollary 3.1.1

Define the *communicability* between two agents as the mutual information between their forecast errors, $C(\rho) := I(\epsilon_i; \epsilon_j)$. When $\rho = 0$, the private components are independent by assumption, so $C(0) = 0$. For $\rho > 0$, agents share the common component ϵ^{pub} , so $C(\rho) > 0$. By the Data Processing Inequality (Shin, 2026), $C(\rho)$ upper-bounds the information any function of ϵ_i (including agent i ’s rationale) can provide to agent j : at $\rho = 0$, deliberation cannot transfer information between agents.

Ensemble MSE decomposition. Our objective is to find $\rho^* = \arg \min_{\rho \in [0,1]} L(\rho)$. The J -agent ensemble MSE decomposes as

$$L(\rho) = \underbrace{\frac{\sigma^2(\rho)}{J} [1 + (J-1)r(\rho)]}_{\text{variance term } V(\rho)} + \underbrace{D(\rho)^2}_{\text{bias term}}, \quad (9)$$

where $\sigma^2(\rho) = \rho^2 \sigma_{\text{pub}}^2 + (1-\rho)^2 \sigma_{\text{priv}}^2$ is the individual agent variance and $r(\rho)$ is the inter-agent correlation from Proposition 3.1. We model $D(\rho)$ as a continuous, non-increasing function of $C(\rho)$, with $D(0) = D_0$ (single-agent bias, since deliberation cannot transfer information when $C(0) = 0$) and $D(\rho) < D_0$ for all $\rho > 0$.

Boundary analysis. At $\rho = 1$: $r(1) = 1$, so $V(1) = \sigma_{\text{pub}}^2$, equal to single-agent variance with no ensemble gain. For any $\rho = 1 - \varepsilon$ with $\varepsilon > 0$: $r(1-\varepsilon) < 1$, so $V(1-\varepsilon) < V(1)$, while $D(1-\varepsilon) \approx D(1)$ by continuity. Therefore $L(1-\varepsilon) < L(1)$, and $\rho = 1$ is not a minimizer.

At $\rho = 0$: $\sigma^2(0) = \sigma_{\text{priv}}^2$, $r(0) = 0$, and $C(0) = 0$. Differentiating, $\left. \frac{d\sigma^2}{d\rho} \right|_{\rho=0} = -2\sigma_{\text{priv}}^2 < 0$ and $\left. \frac{dr}{d\rho} \right|_{\rho=0} = 0$ (the numerator of r vanishes to second order at $\rho = 0$), so

$$\left. \frac{dV}{d\rho} \right|_{\rho=0} = -\frac{2\sigma_{\text{priv}}^2}{J} < 0. \quad (10)$$

By continuity there exists $\varepsilon_V > 0$ such that $V(\varepsilon) < V(0)$ for $\varepsilon \in (0, \varepsilon_V)$; simultaneously, $C(\varepsilon) > 0$ implies $D(\varepsilon) < D_0$. Combining, $L(\varepsilon) < L(0)$ for sufficiently small $\varepsilon > 0$, so $\rho = 0$ is not a minimizer.

Existence of interior optimum. Since neither boundary is a minimizer and $L(\rho)$ is continuous on the compact interval $[0, 1]$, the minimum is attained at some $\rho^* \in (0, 1)$. $\square$

E Experimental Details

E.1 Baseline Descriptions

Single-Agent Methods.

- **Chain-of-Thought** (Wei et al., 2022): one agent receives all evidence and produces a forecast with chain-of-thought reasoning.
- **Self-Consistency** (Wang et al., 2022): the same CoT prompt is sampled 5 times at temperature 0.7 and predictions are averaged.
- **Sequential Bayesian** (Shi et al., 2023): evidence items are processed one at a time by $K=5$ sequential agents, each updating the previous agent’s probability estimate.
- **Superforecaster** (Karger et al., 2024): a structured reasoning prompt that instructs the model to decompose the question, estimate base rates, update with evidence, and perform a calibration check.
- **Halawi et al.** (Halawi et al., 2024): a 7-step structured scratchpad prompt (rephrase, pro/con arguments, aggregate, calibration check) run 3 times, combined via a trimmed mean that downweights the prediction furthest from the median by 50%.

Multi-Agent Methods (Homogeneous Input).

- **Standard Debate** (Du et al., 2024): 3 agents all receive the full evidence corpus, deliberate over 2 rounds sharing predictions and rationales, with final predictions aggregated by mean. As noted in Section 4, our implementation shares InfoDelphi’s prompt and protocol and differs only in information structure.
- **MoA** (Wang et al., 2024): 3 proposer agents independently forecast with full evidence; a 4th aggregator agent receives all proposals with rationales and produces a final synthesized prediction.
- **Crowd Ensemble** (Schoenegger et al., 2024): 5 independent agents each receive all evidence and predict; the median prediction is the final forecast.
- **AIA Forecaster** (Alur et al., 2025): 10 independent agents forecast with full evidence, and a supervisor agent identifies disagreements and produces a final prediction with a confidence label; on low confidence the system falls back to the mean of the 10 agents.

E.2 Computational Cost

A full InfoDelphi run on POLYGYM-250 issues $250 \times 3 \times 2 = 1,500$ forecasting calls plus lightweight aggregation (no additional model calls); baselines range from 250 calls (CoT) to 2,750 (AIA). Cost scales linearly with agents $\times$ rounds.

F Shrinkage Control Experiment

Because the calibrated-shrink transform of Eq. (6) is a post-hoc map on the final probability, it can be granted to any method. Table 6 applies the identical transform (same parameters, no per-method tuning) to every baseline. Three facts emerge. (1) Shrinkage helps every method on both datasets, confirming that the affirmative-side overconfidence it corrects is a model-level bias rather than an artifact of our pipeline. (2) The ranking is preserved: InfoDelphi remains the best system on both datasets. (3) The comparison is honest about effect sizes: with all methods shrunk, several pairwise gaps — notably against Superforecaster, MoA, and Standard Debate on POLYGYM — are no longer individually significant at $n = 250$, although the gaps against 4 of 9 baselines on POLYGYM and 6 of 9 on FutureX remain so. We therefore attribute InfoDelphi’s advantage

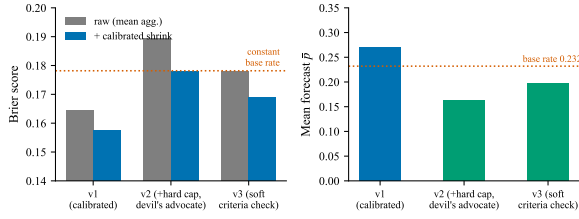


Figure 5: Prompt-level vs. aggregation-level calibration on POLYGYM-250. Left: Brier score of three calibration prompts, with and without aggregation-level shrinkage; the dotted line is a constant base-rate predictor. Right: mean forecast $\bar{p}$ against the empirical base rate. Skeptical prompt wording (v2, v3) depresses the entire forecast distribution below the base rate, whereas asymmetric shrinkage corrects overconfidence where it occurs.

to the combination of information structure and calibrated aggregation, and report both raw and shrunk baselines so the reader can weigh each factor.

G Prompt-Level Calibration Alternatives

An obvious alternative to the aggregation-layer correction of Eq. (6) is to demand skepticism in the prompt itself. We tested two such variants on POLYGYM-250 (prompt text in Appendix J): v2 adds a devil’s-advocate agent and a hard confidence cap; v3 replaces the cap with a soft resolution-criteria checklist. Figure 5 shows that both *hurt*: v2 degrades the raw Brier from 0.1647 to 0.1896 — worse than predicting the constant base rate (0.1782) — and v3 lands at 0.1780. The right panel shows why: prompt-level skepticism cannot be targeted at the overconfident band, so it depresses the whole distribution, pushing the mean forecast from 0.270 down to 0.163 (v2) and 0.198 (v3) against a base rate of 0.232, sacrificing true-YES questions in the mid-band. Shrinkage applied on top of each prompt improves all three variants, but the best combination by a clear margin is the unmodified prompt plus shrinkage (0.1575). Overconfidence corrections belong in the aggregation layer, where they can act asymmetrically on the miscalibrated region, not in the prompt.

H Sports Domain Study

Per-domain analysis shows Sports (106 of 250 POLYGYM questions) is the weakest domain for all methods. A natural hypothesis is missing infor-

mation: match-level evidence often lacks the opponent’s identity or recent form. We tested this with two targeted augmentations: decoding the opponent from the market URL slug (adding it as metadata), and retrieving pre-cutoff news about the decoded opponent’s odds and form under a three-layer leakage control. Both augmentations were consumed — 52–53% of agent forecasts cite the new documents, and 43% of retrieved documents contain numeric odds — yet neither yields a measurable gain (Table 7): paired differences are -0.002 and -0.004 with CIs straddling zero, and untreated control subsets moved as much as treated ones. We interpret sports outcomes as evidence-limited at the *model* level: the residual uncertainty is aleatoric given the kind of evidence a text corpus provides, and no amount of deliberation or retrieval within this class of evidence should be expected to remove it. Information asymmetry improves the *utilization* of informative evidence; it cannot create information the corpus does not contain.

I Significance Tests

Table 8 reports the paired bootstrap intervals behind the daggers in Table 1: for each baseline, the mean question-level Brier difference between InfoDelphi (full) and the baseline, with the 95% percentile interval over 5,000 resamples (seed 0).

J Prompt Templates

All agents receive the same system prompt and structured user prompt; the only difference across agents is their assigned evidence subset.

System prompt (calibrated, v1 — used in all main results).

You are a forecaster for binary (Yes/No) prediction markets. Use only the evidence provided. Prefer high-credibility sources when they conflict.

Follow this procedure:

1. **BASE RATE:** before weighing the evidence, estimate what fraction of questions like this one resolve YES. "Will X happen by <date>" questions usually require a specific chain of events to complete in time, so they resolve NO more often than the surrounding discussion suggests.
2. **EVIDENCE UPDATE:** only concrete, dated evidence that the event is already on track

Table 6: Control experiment: applying our calibrated-shrink transform ($p_0=0.30$, $w_{lo}=0.8$, $w_{hi}=0.5$) to every baseline’s output. Shrinkage improves all methods; the ranking is preserved and InfoDelphi remains best on both datasets, although several pairwise gaps are no longer individually significant (n.s.). Δ is the paired Brier difference of InfoDelphi (full) vs. the shrunk baseline (negative favors ours).

Method	POLYGYM-250			FutureX-231		
	raw	+shrink	Δ [95% CI]	raw	+shrink	Δ [95% CI]
Chain-of-Thought	0.2063	0.1738	−0.016 [−0.029, −0.003]	0.2611	0.2158	−0.015 [−0.033, +0.003] n.s.
Self-Consistency	0.1965	0.1706	−0.013 [−0.025, −0.002]	0.2713	0.2218	−0.021 [−0.038, −0.004]
Sequential Bayesian	0.2523	0.2025	−0.045 [−0.059, −0.030]	0.3043	0.2411	−0.041 [−0.060, −0.021]
Superforecaster	0.1937	0.1665	−0.009 [−0.021, +0.004] n.s.	0.2596	0.2146	−0.014 [−0.031, +0.003] n.s.
Halawi et al.	0.1944	0.1693	−0.012 [−0.024, +0.000] n.s.	0.2654	0.2224	−0.022 [−0.039, −0.004]
Standard Debate	0.1815	0.1650	−0.008 [−0.015, +0.000] n.s.	0.2433	0.2064	−0.006 [−0.022, +0.011] n.s.
MoA	0.1905	0.1670	−0.010 [−0.021, +0.002] n.s.	0.2944	0.2326	−0.032 [−0.049, −0.015]
Crowd Ensemble	0.1992	0.1701	−0.013 [−0.025, −0.000]	0.2888	0.2309	−0.030 [−0.048, −0.012]
AIA Forecaster	0.1897	0.1651	−0.008 [−0.019, +0.004] n.s.	0.2764	0.2223	−0.022 [−0.039, −0.003]
InfoDelphi (full)	0.1647	0.1575	—	0.2212	0.2006	—

Table 7: Targeted retrieval augmentation on the 106 Sports questions of POLYGYM-250 (InfoDelphi full; paired bootstrap vs. original evidence, 5,000 resamples). Neither augmentation yields a significant improvement.

Evidence	Brier ↓	AUC ↑	Δ Brier [95% CI]
Original	0.1908	0.696	—
+ opponent identity	0.1890	0.690	−0.0018 [−0.010, +0.006]
+ opponent odds/form	0.1873	0.720	−0.0035 [−0.013, +0.005]

should move you materially above the base rate. Topical coverage, speculation, or enthusiasm is not evidence for YES.

3. SHRINK: if the evidence is indirect, incomplete, or stale, move your estimate toward the base rate -- not toward 0.5.

4. CALIBRATION CHECK: out of 100 questions where you would give this exact probability, how many should actually resolve YES? Adjust if your number feels like a hedge or a headline.

You must respond with valid JSON only, no other text before or after.

Prompt variants of the calibration study (Appendix G). Variant v2 inserts a *resolution-criteria* step and a *high-confidence gate*: before giving $p_{yes} \geq 0.7$, the agent must verify every resolution criterion against cited, dated evidence and otherwise “stay at or below 0.6”; it additionally runs a devil’s-advocate agent. Variant v3 removes the hard cap and instead appends a soft re-check to the calibration step: when any resolution criterion is supported only by analogy, “move back toward the base rate.”

User prompt (round 1).

```
Question: {question}
Evidence:
[1] doc_id=doc_001 | source=...
| title=... | content: ...
[2] doc_id=doc_002 | source=...
| title=... | content: ...
...
Respond with exactly one JSON
object, no other text. Schema:
{"p_yes": <0-1>, "label":
"YES"|"NO", "rationale": "...",
"evidence_used": ["doc_id",
...]}
```

User prompt (round 2). The round-2 prompt appends a history summary of all other agents’ round-1 outputs after the evidence:

```
Previous round summary (other
agents):
Agent 1: p_yes=0.850 (YES) --
The evidence suggests...
Agent 2: p_yes=0.120 (NO) --
Based on the pricing data...
Round mean p_yes=0.533.
Consider whether you agree or
disagree with the above.
```

With evidence pooling enabled, rationales cite documents by doc_id together with the concrete fact each contributes, so privately-held documents referenced in a rationale become visible context for other agents in the next round.

Table 8: Paired bootstrap comparison of InfoDelphi (full) against each baseline (5,000 resamples over questions, seed 0). $\Delta\text{Brier} < 0$ favors InfoDelphi; all 95% CIs exclude zero.

Baseline	POLYGYM-250 ΔBrier [95% CI]	FutureX-231 ΔBrier [95% CI]
Chain-of-Thought	-0.0488 [-0.0756, -0.0217]	-0.0606 [-0.0958, -0.0258]
Self-Consistency	-0.0390 [-0.0619, -0.0149]	-0.0707 [-0.1032, -0.0381]
Sequential Bayesian	-0.0948 [-0.1209, -0.0681]	-0.1037 [-0.1402, -0.0684]
Superforecaster	-0.0362 [-0.0620, -0.0097]	-0.0591 [-0.0941, -0.0248]
Halawi et al.	-0.0370 [-0.0615, -0.0123]	-0.0648 [-0.0966, -0.0338]
Standard Debate	-0.0240 [-0.0416, -0.0064]	-0.0427 [-0.0761, -0.0114]
MoA	-0.0331 [-0.0573, -0.0086]	-0.0938 [-0.1295, -0.0580]
Crowd Ensemble	-0.0418 [-0.0662, -0.0167]	-0.0882 [-0.1224, -0.0534]
AIA Forecaster	-0.0323 [-0.0566, -0.0077]	-0.0758 [-0.1103, -0.0409]